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Obstetric Anesthesia Workforce Survey: Forty-year Update

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EDITOR'S PERSPECTIVE

What We Already Know about This Topic

- Obstetric Anesthesia Workforce Surveys have been conducted every decade for the past 40 yr
- Each survey has assessed trends in obstetric anesthesia practice and identified potential areas for improvement

What This Article Tells Us That Is New

- Despite stable annual birth rates in the United States, the number of hospitals providing obstetric care has decreased by 50% over the last 40 yr
- Access to neuraxial labor analgesia has dramatically increased
- Use of neuraxial labor analgesia has increased in all size hospitals over time, associated with lower rates of general anesthesia for cesarean deliveries

ABSTRACT

Background: For 40 yr, Obstetric Anesthesia Workforce Surveys have been used every decade to assess trends in obstetric anesthesia practice and potential areas for improvement. Anesthesia providers from U.S. hospitals were surveyed in 2022 to 2023 to provide data for their hospitals from 2021. The primary hypothesis was that obstetric anesthesia services have changed in the last decade.

Methods: Previous workforce surveys were used to develop the 32-question survey about contemporary obstetric anesthesia practice. A hospital sample ($n = 1,180$) was generated based on number of births per year and U.S. census region. Using web-push survey methodology, a QR code was assigned to the "Chief of Anesthesiology" at each hospital. A link to an online Research Electron Data Capture survey was emailed to individuals along with reminder communications. Nonresponding hospitals received paper surveys and self-addressed stamped envelopes for survey return. The results were analyzed using R statistical package at a significance level of $P < 0.05$.

Results: There were 284 (24%) responses to the survey. Hospitals providing obstetric care have decreased 50% over four decades. Of all the respondents, 77% work in nonacademic hospitals without residency programs. Comparing academic to nonacademic hospitals, academic providers are less likely to have other clinical responsibilities: 35% *versus* 62%, respectively. The weighted overall rate of neuraxial labor analgesia is 84%. Elective cesarean deliveries are usually performed with spinal anesthesia (85%). Neuraxial anesthesia is used in 86% of urgent cesareans, while 14% use general anesthesia.

Conclusions: Despite stable annual birth rates in the United States, the number of hospitals providing obstetric care decreased by 50% over the last 40 yr. This study describes nonacademic practice and not just academic teaching hospitals. Increased access to neuraxial labor analgesia is a dramatic step toward reducing in-hospital maternal mortality and improving healthcare disparities.

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“Obstetric anesthesia, if it would just go away, we would all be happy.”¹ This quote was published in 1984 by a state anesthesiology society president before

the first obstetric anesthesia workforce survey publication in 1986.² The reasons for this sentiment have not changed: unpredictability, life-threatening emergencies, required

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Abbreviations: ACLS, Advanced Cardiac Life Support; PCEA, patient-controlled epidural analgesia; REDCap, Research Electron Data Capture; SOAP, Society for Obstetric Anesthesia and Perinatology

communication skills in dynamic environments, immediate need for general anesthesia, staffing challenges, insufficient payment for services provided, awake patients, and families. Practice guidelines and consensus statements have helped to address anesthesia-related issues, but there are major concerns now outside anesthesiology, including rising non-anesthesia-related maternal mortality rates, care disparities, reproductive care challenges, and maternal care deserts.^{3,4} Since the 1986 publication, U.S. obstetric anesthesia workforce surveys have been used every 10 yr to describe current practice and identify areas needing improvement.^{2,5–7} This publication is unique because it longitudinally documents 40 yr of progress in obstetric anesthesia.

Earlier obstetric anesthesia workforce surveys allow for reflection on how clinical care has changed over four decades. In 1981, 30% of labor epidurals and 42% of spinal anesthetics were administered by obstetricians.² One conclusion from survey findings was “availability of obstetric anesthesia personnel needs improving.” Anesthesiologists heeded the call, and by 1991, only 2% of labor epidurals were reported as placed by obstetricians,⁵ and general anesthesia for obstetric cases had also decreased. However, up to a third of patients received no labor analgesia. By 2001, the availability of anesthesia care had dramatically increased, with 95% of the smallest hospitals stating that obstetric anesthesia care was available on-call and 80% of larger hospitals reporting in-house availability.⁶ Patient-controlled epidural analgesia (PCEA) was used in up to 35% of laboring patients. By 2011, up to 80% of patients received neuraxial labor analgesia, PCEA use doubled, and there was 100% availability of anesthesia care in the largest hospitals.⁷ Questions were also introduced about emergency practices and postpartum tubal ligations in 2011.

Since that publication in 2016,⁷ there have been substantial advances in education and clinical practice within and outside the subspecialty. Collaborations with other societies have produced consensus statements and guidelines by the Society for Obstetric Anesthesia and Perinatology (SOAP; Lexington, Kentucky), as well as the American Society of Anesthesiologists (Schaumburg, Illinois).^{8–10} Accreditation Council for Graduate Medical Education-accredited fellowships have been developed in obstetric anesthesia, and anesthesiologists are members of state maternal mortality review committees. In addition, the SOAP Center of Excellence designation recognizes institutions that meet peer-reviewed criteria for excellence in obstetric anesthesia care.¹¹ To match parturients' needs with centers resourced to care for them, the American College of Obstetricians and Gynecologists (Washington, D.C.) developed a Maternal Level of Care Consensus Statement.¹² These levels of care were adopted by the Joint Commission¹³ and contribute to advancing care and safety of obstetric patients. Our primary hypothesis was that aspects of labor and delivery care had changed in the last 10 yr. This study reports results from a 2022 survey of

obstetric anesthesia providers from a representative sample of U.S. hospitals with labor and delivery units.

Materials and Methods

We used 2020 data from the American Hospital Association's (Chicago, Illinois) Guide to the Health Care Field, (2021 edition)¹⁴ on all 2,846 U.S. hospitals with one or more births. A random sample of 1,180 U.S. hospitals was stratified by 9 U.S. Census Bureau Regions and Divisions (New England, Mid Atlantic, South Atlantic, East North Central, East South Central, West North Central, West South Central, Mountain, and Pacific)¹⁵ and 5 delivery size groups (more than 4,500 births, 3,000 to 4,499 births, 1,500 to 2,999 births, 500 to 1,499 births and fewer than 500 births). An internet search for the “chief of anesthesiology” was performed for each hospital. If an individual was identified, the American Society of Anesthesiologists member directory was searched for their email address. If not, sampled hospitals were contacted by telephone to determine the name and email address of the chief of obstetric anesthesiology, an obstetric anesthesiology provider, or the chief of anesthesiology. The searches included SOAP members, obstetric anesthesia division chiefs, obstetric fellowship directors, and personal knowledge of obstetric anesthesiologists by the survey team at individual hospitals that were part of the random sample. If contact information was obtained, a survey invitation with a unique link to an online Research Electronic Data Capture (REDCap)¹⁶ survey was emailed to the identified individual. If contact information was not obtained, a survey invitation with a unique QR code linked to the hospital was sent *via* postal mail to the attention of the chief of anesthesiology at the hospital.

Survey invitations described the purpose and importance of the study and provided contact information for study personnel in case sample members had questions about the survey or needed to share the name of another provider at the hospital who was more appropriate for the survey. Initial email/postal letter invitations were followed by three reminder communications (emails/postal letters) to nonresponding hospitals. Following a push-to-web methodologic approach,¹⁷ a final mailing including a reminder letter, a paper copy of the survey, and a stamped return envelope was sent to all hospitals that did not respond after all reminders. Web-push data collection methods have become a standard of practice that is widely used in survey research, including in large-scale government surveys in the United States and around the world. This methodology has been associated with improved coverage and response rates, as well as reduced survey costs. There is also growing evidence that a push-to-web approach is effective at improving response to physician surveys while reducing costs and fielding time.¹⁷

The survey included 32 questions (appendix 1) about the type and number of obstetric anesthesia providers

providing labor and delivery services, staffing patterns, provider availability, midlevel provider supervision, availability of emergency training for all labor and delivery staff, use of hemorrhage protocols, type of anesthesia administered for cesarean delivery, vaginal delivery, and postpartum tubal ligation, cesarean delivery rate, the availability of trial of labor after cesarean, and its impact on staffing. The survey was in the field between January 30, 2022, and September 1, 2023. All survey procedures were approved by the Colorado Multiple Institutional Review Board (Aurora, Colorado; approval No. 22-0970).

All statistical analyses were performed at the significance level of 0.05 and analyzed using R (version 4.3.1). To account for survey nonresponse, Horvitz–Thompson estimators were incorporated, weighing responses by the inverse of the probability of response.¹⁸ Stratum-specific weights were used to calculate response estimates overall from and among delivery size-based strata. The Wilson procedure with a correction for continuity was used to calculate weighted 95% confidence limits. Chi-squared testing was used to test for differences in proportions across mentioned delivery size-based stratum. We additionally incorporated delivery size-based stratum from previous surveys to compare estimates across time. To account for item missingness, we ran a sensitivity analysis in which we assumed all missing items were coded as “yes” in the surveys before rerunning the Horvitz–Thompson analyses. We descriptively compared clinical responsibilities while covering labor and delivery, as well as stipend provision for in-house labor and delivery between academic and community hospitals. These numbers were calculated without weights to facilitate overall comparisons instead of comparisons across strata.

Results

Number of Hospitals Providing Obstetric Care

Table 1 represents the number of hospitals providing obstetric care by the annual number of births from 1981 to 2023.

U.S. Hospitals Surveyed and the Births They Represent

Table 2 contains data aggregated from the American Hospital Association’s Annual Survey Database for the fiscal year 2020 by strata. Overall, 2,846 hospitals in the United States provide obstetric care for greater than 3.5 million deliveries. For this report, the data are divided into strata based on how many deliveries occur in their hospital per year, with stratum 1 representing those hospitals with the largest delivery services of greater than 4,500 deliveries per year, stratum II with 3,000 to 4,499 deliveries, stratum III with 1,500 to 2,999 deliveries, stratum IV with 500 to 1,499 deliveries, and stratum V with the smallest labor and delivery services of less than 500 deliveries. Appendix 2 provides comprehensive response

Table 1. Number of U.S. Hospitals Providing Obstetric Care by Annual Number of Births

Survey Year	Hospitals with Obstetric Care	Total Births
1981	4,163	3,667,316
1991	3,545	3,669,112
2001	3,160	3,973,925
2011	2,900	3,888,844
2020	2,846	3,536,150
2022*	2,075	3,661,220
2023	1,893	3,591,328

*2022 data were used to assess postpandemic hospital closures and birth rates.

rates from 1981 to 2021 surveys. For this survey, 1,180 hospitals were contacted, and 284 responses were received for a 24% survey response rate. The 284 responding hospitals represented about 12% of deliveries in the United States in 2020.

Types of Anesthesia Practices and Responding Hospitals

Table 3 describes type of anesthesia practice by the size of their hospital delivery service. Each of the categorical variables is represented as an estimated weighted N (%) of responding hospitals. Overall, 77% of respondents describe their practice as nonacademic without residents, but that ranges from 45% in stratum I to 79% in stratum V. This is statistically significantly different ($P < 0.01$) between strata. The next most common type of practice was academic with residents for 14% of respondents, primarily at hospitals with services more than 1,500 deliveries per year. Respondents from military and Indian Health Services hospitals reported smaller delivery services, mainly in strata IV and V. When asked whether their hospitals were regional referral centers for high-risk obstetrics, 100% of stratum I hospitals responded “yes”; this percentage decreased as size of delivery service fell to only 9% in stratum V hospitals.

Staffing Models Used on Labor and Delivery

Table 4 shows data on staffing models used on labor and delivery during weekday, night, and weekend hours by size of hospital delivery service. Like in table 3, each of the categorical variables is represented as an estimated weighted N (%) of responding hospitals. Overall, an estimated weight of responding hospitals (47%) reported a care team model with a physician anesthesiologist supervising residents, certified registered nurse anesthetists, or anesthesiologist assistants for weekday coverage. Anesthesiologists personally provide care in about 23% of hospitals overall, resulting in an anesthesiologist being present on labor and delivery in the majority of hospitals. As delivery services become smaller, more were reported to be staffed by a certified

Table 2. U.S. Hospitals Surveyed and the Births They Represent

Strata	Hospitals in Stratum (N)	Hospitals Surveyed		Births N	Births Surveyed	
		N	%		N	%
Stratum I: >4,499 births	108	22	20	704,377	127,509	18
Stratum II: 3,000–4,499 births	171	30	18	624,475	104,729	17
Stratum III: 1,500–2,999 births	494	52	11	1,053,957	92,807	9
Stratum IV: 500–1,499 births	956	87	9	874,700	76,975	9
Stratum V: <500 births	1,117	93	8	278,641	21,356	8
Total N	2,864*	284†	10	3,536,150	423,376	12

The table shows data aggregated from the American Hospital Association's Annual Survey Database for the fiscal year 2020.

*All U.S. hospitals with one or more births. †Of 1,180 hospitals in the random sample, there were 284 survey responses.

registered nurse anesthetist without physician supervision, including 66% of the smallest stratum V delivery services. Similar staffing patterns were seen on labor and delivery at night and on weekends.

Respondents at many hospitals report that anesthesiologists are covering other clinical responsibilities in addition to labor and delivery, even in those hospitals with the largest delivery services. In stratum I hospitals, those having other clinical responsibilities ranged from 11% during daytime hours to 19% during nights and weekends. When comparing academic to nonacademic hospitals, academic providers are less likely to be covering other clinical responsibilities: 35% *versus* 62%, respectively. However, during nights and weekends, academic practices with other clinical responsibilities increase to 50%, while nonacademic practices stay about the same.

Types of Analgesia Provided during Labor

Table 5 summarizes the responses and compresses the data to three strata to allow comparison with previous surveys. In this table, stratum I is all delivery services with more than 1,500 deliveries, stratum II is 500 to 1,499 deliveries, and stratum III is less than 500 deliveries. Respondents were asked to make their best estimate of what percentage of patients used various forms of pain relief during labor. Responses could add up to more than 100% because patients may have used more than one modality during labor. The current data show no statistically significant differences ($P < 0.05$ considered significant) by strata for unmedicated births using no analgesia ($P = 0.36$), for intravenous analgesics ($P = 0.12$), and for single-injection spinal analgesia ($P = 0.26$). The estimated overall percentages for each of these types of analgesia were 10, 21, and 6%, respectively (see table 5). *Via* a chi-square test, use of epidural analgesia does not significantly vary between strata ($P = 0.03$). Figure 1 summarizes the rates of labor epidural utilization and use of general anesthesia for cesarean delivery in all sizes of hospitals per year from 1981 to 2021. Figure 2 summarizes the rates of labor epidural utilization and use of general anesthesia for

cesarean delivery in hospitals with less than 500 deliveries per year from 1981 to 2021. The weighted overall rate of neuraxial analgesia for labor is 84%. The most common epidural maintenance technique is continuous infusion plus PCEA, used by 73%. In contrast, programed intermittent epidural bolus with PCEA is used by 15% overall, most commonly in the stratum I and II hospitals. The data for nitrous oxide utilization can be found in table 5.

Trial of labor after cesarean is allowed in 83% of hospitals overall, ranging from 100% in stratum I to 48% in stratum V. For hospitals who allow trial of labor after cesarean, greater than 90% require the in-house presence of an obstetrician in stratum I to IV hospitals but only 62% in stratum V hospitals. A certified registered nurse anesthetist or anesthesiologist is required to be in-house in 69% of hospitals overall for trial of labor after cesarean.

Anesthetic Techniques Used for Cesarean Delivery

Table 6 describes the types of anesthesia used for elective and urgent cesarean deliveries, using the compressed strata for comparison to previous surveys. The results are very similar across hospitals with different sizes of delivery services. Urgent cesarean deliveries are most commonly done with neuraxial anesthesia, using an epidural placed for labor if present or a spinal anesthetic if a new anesthetic is initiated. General anesthesia is used for 14% of urgent cesarean deliveries overall. Figure 1 demonstrates that rates of general anesthesia have decreased as epidural labor analgesia utilization has increased in all sizes of hospitals. Across strata, spinal anesthesia was used most often for elective cesarean deliveries (85% overall). Data on use of neuraxial morphine for postoperative analgesia and enhanced recovery protocol for cesarean delivery can be found in table 6.

Provision of Postpartum Tubal Ligation Surgeries

Survey recipients were asked whether their hospital provides postpartum tubal ligation surgeries during the

Table 3. Description of Hospitals within Each Birth Stratum in the United States from 2021

Strata	Number of Hospitals Surveyed	Number and Percentage of Regional Referral Centers for High-risk Obstetrics	Academic Practice				Nonacademic Practice			
			With Anesthesiology Residency Program		Without Anesthesiology Residency Program		Private Practice with Anesthesiology Residents		Practice without Anesthesiology Residents	
			N	%	N	%	N	%	N	%
Stratum I: >4,499 births	49	49	19	39	4	8	2	4	22	45
Stratum II: 3,000–4,499 births	69	51	28	41	1	1	6	1	33	48
Stratum III: 1,500–2,999 births	213	116	54	25	22	10	11	5	109	51
Stratum IV: 500–1,499 births	387	111	41	11	32	8	4	1	261	67
Stratum V: <500 births	444	39	19	4	14	3	12	3	349	79
Total N	1,162*	366	161		73		35		774	

The table shows data gathered in 2022 to 2023 reflecting practice in 2021. The breakdown of each categorical variable is represented as estimated weighted N (%) of responding hospitals within each birth stratum.

*Compared to the random sample of $n = 1,180$, $n = 1,162$ is based on the number of hospitals for which we have data per question per strata and the number of American Hospital Association hospitals per strata. This missingness does not necessarily follow the same pattern across strata for each question.

delivery admission, whether anesthesia staffing affects the ability to perform the surgery, and what type of anesthesia is used for the procedure. Most hospitals perform postpartum tubal ligation during the delivery admission (74% overall) with no statistically significant differences across strata ($P = 0.65$). In stratum I hospitals, 29.7% (95% CI, 12.3 to 55.96) responded that anesthesia staffing never affected performance of postpartum tubal ligation, while 50.9% (95% CI, 38.42 to 63.35) of the smallest stratum V delivery services said that anesthesia staffing was never an issue. There was a statistically significant difference between the distribution of proportions of how often inadequate anesthesia staffing affects the ability to perform the surgery across strata ($P < 0.01$). Table 7 describes anesthetic techniques used for postpartum tubal surgeries.

Patient Safety Initiatives on Labor and Delivery

Respondents were asked several questions related to patient safety initiatives and resources available to them on labor and delivery, and their responses are summarized in table 8. Most delivery services reported having a means to provide antenatal consults, although larger delivery services were more likely to have a formal clinic or antenatal consultation service, while smaller delivery services were more likely to provide consults on an *ad hoc* basis. For example, 60.4% (95% CI, 49.1 to 70.71) of hospitals in stratum V provide consults on an *ad hoc* basis compared to only 6.4% (95% CI, 2.67 to 14.44) who have a clinic or consultation service. In contrast, only 34.2% (95% CI, 15.18 to 60.09) of hospitals in stratum I provide consults on an *ad hoc* basis compared to 57.9% (95% CI, 33.53 to 78.9) who have a formal clinic or antenatal consultation service. Most hospitals of all sizes reported having a massive transfusion protocol (92% overall). Availability of point-of-care testing was very dependent on size of delivery service, ranging from 83% in stratum I hospitals to 30% in stratum V hospitals. The proportion of hospitals that have point of care testing was statistically significantly different across strata ($P < 0.01$). There were statistically significant differences in the percentages of hospitals in each stratum ($P < 0.01$) that required Advanced Cardiac Life Support (ACLS) certification for nurses on labor and delivery, but nearly 25% of respondents overall did not know whether their labor and delivery nurses were ACLS certified. Most hospitals have a rapid response team that can respond to critical events on labor and delivery (83% overall), although it was less available in the smallest delivery services at only 70%. There was a statistically significant difference in the proportion of hospitals that had a rapid response team across strata.

Ultrasound was widely available for neuraxial procedures, either dedicated to the anesthesia service in 44% overall or borrowed from the obstetric or pain services in 22%. In contrast, most delivery services did not have staff proficient with the acquisition and interpretation

Table 4. Staffing Models Used on Labor and Delivery in the United States from 2021

Staffing Models	Stratum I: I: >4,499 Births		Stratum II: 3,000–4,499 Births		Stratum III: 1,500–2,999 Births		Stratum IV: 500–1,499 Births		Stratum V: <500 Births		Overall	
	N	%	N	%	N	%	N	%	N	%	N	%
Weekdays												
Care team model with physician anesthesiologist supervision	37	74	55	78	132	62	238	62	92	20	554	47
CRNA without physician supervision	0	0	2	3	16	7	37	10	301	66	56	30
Physician anesthesiologist providing direct care	13	26	13	19	66	31	112	29	62	13	265	23
When covering L&D, do physician anesthesiologists have other clinical responsibilities?	6	11	25	36	123	58	248	67	268	76	670	63
Nights and weekends												
Care team model with physician anesthesiologist supervision	35	70	57	81	125	59	190	49	72	16	478	41
CRNA without physician supervision	0	0	0	0	16	8	53	14	314	69	383	33
Physician anesthesiologist providing direct care	15	30	13	19	70	33	143	37	68	15	308	26
When covering L&D, do physician anesthesiologists have other clinical responsibilities?	9	19	38	55	155	73	264	71	266	75	733	69

The table shows data gathered in 2022 to 2023 reflecting practice in 2021. The breakdown of each categorical variable is represented as estimated weighted N (%) of responding hospitals within each birth stratum.

L&D, labor and delivery; CRNA, certified registered nurse anesthetist.

Table 5. Techniques Used for Labor Analgesia 1981 to 2021

Technique	Stratum I					Stratum II					Stratum III				
	1981	1991	2001	2011	2021	1981	1991	2001	2011	2021	1981	1991	2001	2011	2021
None, %	27	11	6	9	8	33	14	10	9	11	45	33	12	14	10
Parenteral, %	52	48	34	25	19	53	60	42	40	23	37	48	37	34	21
Paracervical block, %	5	2	2	2	2	5	4	3	7	1	6	7	3	3	0
Neuraxial (all types), %	22	51	69	86	89	13	33	50	85	83	9	17	40	65	84
Single-injection spinal, %	0	4	8	NA	5	0	4	9	NA	7	0	4	17	NA	6
Epidural, %	22	51	61	71	61	13	33	42	67	68	9	17	35	45	64
Dural puncture epidural, %	—	—	—	—	10	—	—	—	—	2	—	—	—	—	4
CSE, %	—	—	8	15	10	—	—	8	18	7	—	—	5	16	6
Nitrous oxide, %	—	—	—	—	11	—	—	—	—	4	—	—	—	—	3

The data represent the estimated percentages of responses and are compressed to three strata to allow comparison to previous surveys. The table shows the techniques used for all types of labor analgesia. Parturients may have received more than one technique. A dash indicates that the question was not asked in this survey.

CSE, combined spinal epidural; NA, data not available.

of transthoracic echocardiography images. Transthoracic echocardiography was available in about half of stratum I and II hospitals and in only 7% of stratum V hospitals. These percentages were statistically significantly different across strata ($P < 0.01$). Anesthesiologists rarely have a role in newborn resuscitation in any size hospital, although it is more common in the smallest delivery services, with 15% responding “yes” in stratum V. In contrast, 66% overall said they will respond in a neonatal emergency, and this was similar across strata, although the percentages were statistically significantly different across strata ($P = 0.03$).

Appendix 3 summarizes the financial and patient demographic information by strata.

Discussion

This fifth workforce survey of U.S. obstetric anesthesia practice builds on previous publications during the four decades since 1981.^{2,5–7} Current survey results are notable because 77% of responding institutions were nonacademic hospitals without residency teaching programs, indicating that this study reflects community practice and not just academic teaching hospitals. Since most studies of obstetric anesthesia practice focus on academic program directors, chairs, or division chiefs, the results of this survey are unique.^{19–21} The strength of this and previous workforce surveys is the effort made to contact anesthesiologists in all types of practice environments. To our knowledge, it is the only longstanding

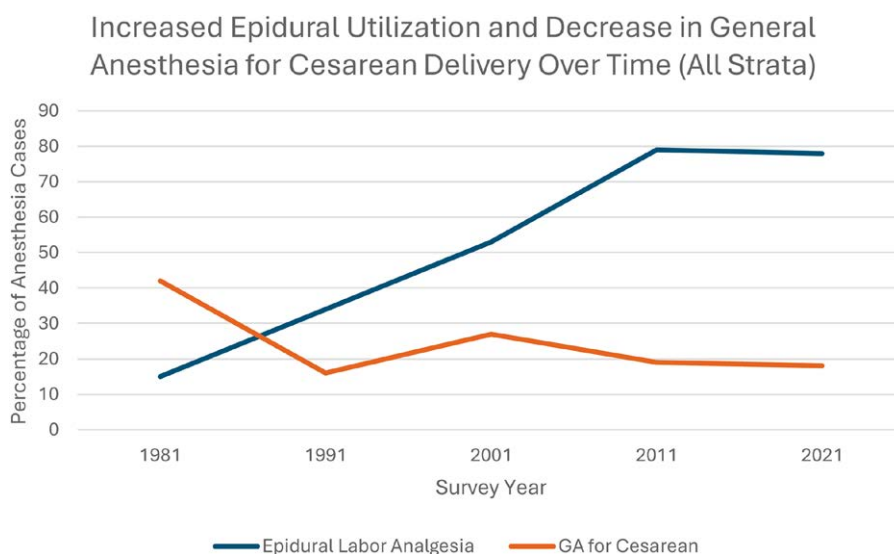


Fig. 1. Rates of labor epidural utilization and decreased use of general anesthesia for cesarean delivery over time in all strata. GA, general anesthesia.

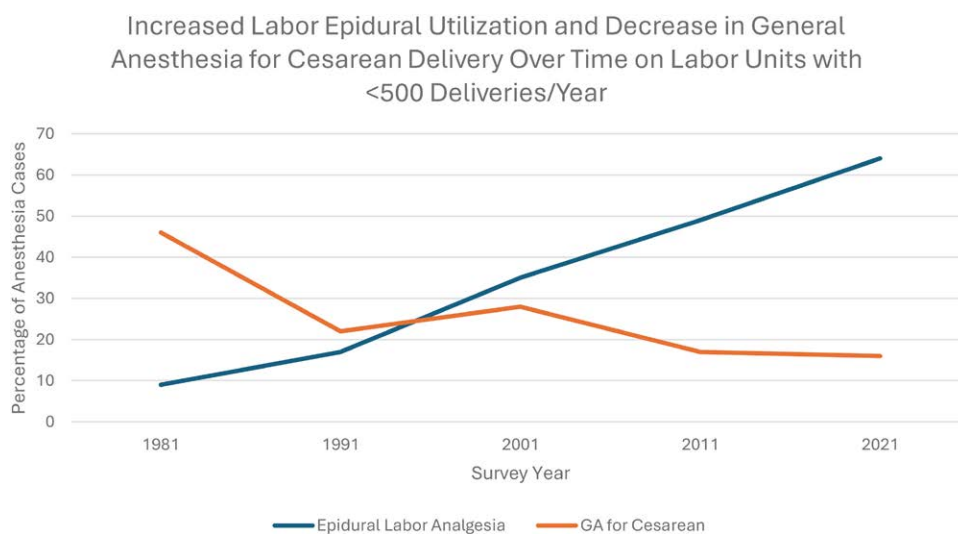


Fig. 2. Rates of labor epidural utilization and use of general anesthesia for cesarean delivery in hospitals with less than 500 deliveries per year over time. GA, general anesthesia.

survey that provides a snapshot of U.S. obstetric anesthesiology practice each decade since 1981.

The initial survey was undertaken in 1981 because of widespread problems with anesthesia staffing on labor and delivery.² Since 1981, there has been improvement in availability of obstetric anesthesia services. At the same time, the number of U.S. births has remained fairly constant, while the number of hospitals providing obstetric care has decreased by almost half, raising concern that maternity care is inaccessible in many areas (see table 1), especially

since the COVID-19 pandemic.³ The term “maternity care desert” is used to describe a county without a hospital or birth center that offers obstetric care or without any obstetric providers.^{3,4} In the current survey, most respondents report obstetric anesthesia services are available 24/7. The care team was the most common staffing model. Similar staffing models were seen on labor and delivery at night and on weekends. However, many respondents report having additional duties beyond labor and delivery, even in hospitals with the largest delivery volume, and

Table 6. Anesthetic Techniques for Cesarean Delivery 1981 to 2021

Technique	Stratum I					Stratum II					Stratum III				
	1981	1991	2001	2011	2021	1981	1991	2001	2011	2021	1981	1991	2001	2011	2021
GA, %	35	12	—	—	—	45	15	—	—	—	46	22	—	—	—
Urgent	—	—	15	19	16	—	—	30	17	14	—	—	25	14	13
Elective	—	—	3	3	2	—	—	5	4	3	—	—	3	3	3
Epidural, %	29	54	—	—	—	16	45	—	—	—	12	29	—	—	—
Urgent	—	—	36	43	53	—	—	21	41	55	—	—	14	34	50
Elective	—	—	22	15	5	—	—	17	13	4	—	—	14	16	2
Spinal, %	33	35	—	—	—	35	40	—	—	—	37	49	—	—	—
Urgent	—	—	45	41	25	—	—	48	56	28	—	—	59	61	39
Elective	—	—	67	75	77	—	—	75	88	86	—	—	80	87	93
CSE, %	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Urgent	—	—	4	12	6	—	—	2	5	3	—	—	2	5	1
Elective	—	—	8	17	16	—	—	2	9	7	—	—	3	4	0
Neuraxial morphine, %	—	—	—	—	90	—	—	—	—	86	—	—	—	—	78
ERAS protocol, %	—	—	—	—	76	—	—	—	—	53	—	—	—	—	48

The data represent estimated percentages of responses and are compressed to three strata to allow comparison to previous surveys. A dash indicates that the question was not asked in the surveys.

CSE, combined spinal epidural; ERAS, Enhanced Recovery After Surgery; GA, general anesthesia.

Table 7. Anesthetic Techniques Used for Postpartum Tubal Ligation Surgery in the United States from 2021

Anesthetic Techniques	Stratum I: >4,499 Births	Stratum II: 3,000–4,499 Births	Stratum III: 1,500–2,999 Births	Stratum IV: 500–1,499 Births	Stratum V: <500 Births	Overall
Spinal, %	50	53	36	36	15	32
Epidural from labor, %	39	11	30	14	34	26
General anesthesia, %	8	34	34	50	35	37

The tables shows data gathered in 2022 to 2023 reflecting practice in 2021. The numbers represent the weighted estimate average percentage reported by the surveyed hospitals.

these additional duties increase substantially at night and on weekends. Although some aspects of obstetric care are predictable (*e.g.*, scheduled cesarean delivery), the labor and delivery workload is generally nonlinear, with requests for labor analgesia and emergencies occurring randomly throughout the day and night.^{22,23} This is an important area of emphasis in our results since multiple studies report higher rates of adverse outcomes including lower umbilical cord pH, higher cesarean delivery rate, lower neonatal Apgar scores, and perinatal death when deliveries occur at night or on the weekend.^{24,25} Given that our respondents reported having less staff and additional responsibilities during these times, this is an area needing improvement. Ginosar *et al.*²³ caution that maternity units, operating rooms, emergency rooms, and intensive care units often operate at “maximum capacity or even above it” and that “it is the workload variability and not the average workload, that must be the driver for margins of safety.” Further study is needed to determine whether these reported decreases in staffing have an impact on these outcomes. One contributing factor to decreases in staffing at night and on weekends might be that only 27% of respondents

reported a stipend from the hospital to offer services outside of regular business hours.

In the 1990s, anesthesiologists focused on reducing airway-associated morbidity and mortality related to obstetric anesthesia.⁵ Spinal anesthesia is still the most common anesthetic for cesarean delivery. Rates of general anesthesia for cesarean delivery have remained relatively constant since 2001, especially in stratum I hospitals where respondents report approximately 2% for elective and 16% for urgent cesarean delivery. The recommended general anesthesia rate for a hospital to be considered for the Center of Excellence designation by SOAP is 5% or less.¹¹ Given that rates of general anesthesia greater than 5% are consistent across strata and over time, studies should be undertaken to determine factors that affect general anesthesia utilization and whether the recommended rates of general anesthesia¹¹ should be modified. In accordance with SOAP Center of Excellence recommendations, most hospitals provide neuraxial morphine for postcesarean analgesia and have an enhanced recovery protocol.

Use of neuraxial labor analgesia has increased over time across all strata, associated with lower rates of general

Table 8. Patient Safety Initiatives Used on Labor and Delivery in the United States from 2021

Safety Initiative	Responses	Stratum I: >4,499 Births		Stratum II: 3,000–4,499 Births		Stratum III: 1,500–2,999 Births		Stratum IV: 500–1,499 Births		Stratum V: <500 Births		Overall	
		N	%	N	%	N	%	N	%	N	%	N	%
Is antenatal consultation for high-risk patients available before arrival in L&D?	Yes	29	58	39	56	79	37	95	25	29	6	270	23
	Yes, but only on an <i>ad hoc</i> basis	17	34	23	32	112	52	221	57	275	60	647	55
Does your hospital have a massive transfusion protocol?	Yes	49	100	66	94	214	100	367	95	389	85	1,085	92
Is point-of-care testing available in L&D?	Yes	41	83	59	85	173	89	199	52	137	30	610	52
Does your hospital require ACLS certification for L&D nurses?	Yes	25	51	46	66	101	48	214	56	341	74	727	62
Does your hospital have a rapid response team that responds to critical events in L&D?	Yes	44	95	59	87	202	94	352	91	320	70	977	83
Do you have access for neuraxial ultrasound L&D?	Yes, borrowed from obstetrician or pain service	7	15	7	10	65	30	53	14	131	29	263	22
	Yes, dedicated to the anesthesia service	31	63	43	62	98	46	169	44	182	40	524	44
Do you have anesthesiology staff proficient with acquisition and interpretations of TTE images for L&D?	Yes	23	48	41	58	106	50	74	19	32	7	276	23

The table shows data gathered in 2022 to 2023 reflecting practice in 2021. The breakdown of each categorical variable is represented as estimated weighted N (%) of responding hospitals within each birth stratum.

ACLS, Advanced Cardiac Life Support; L&D, labor and delivery; TTE, transthoracic echocardiogram.

anesthesia for cesarean (figs. 1 and 2). In the United States, the weighted overall prevalence of neuraxial analgesia for labor is now 84%. Although 40 yr ago, 77 to 80% of parturients across all strata received no analgesia or parenteral opioids, this has decreased substantially to only 10% in 2021. It is important to emphasize that use of neuraxial analgesia has increased in all strata, with stratum III to V hospitals reporting similar utilization to larger hospitals for the first time since the inception of these surveys. This appears to reflect increased labor analgesia access in smaller hospitals, which may help reduce maternal morbidity and mortality.³ Increased epidural utilization is critically important for improving outcomes such as general anesthesia for cesarean delivery, which has been reported to have higher rates of morbidity and mortality.²⁶ Our longitudinal data (figs. 1 and 2) demonstrate that as neuraxial labor analgesia utilization rates have increased, the rate of general anesthesia for cesarean delivery has decreased. Higher rates of general anesthesia for cesarean delivery have been reported in Black and Hispanic parturients, but a recent study demonstrated that this disparity is mitigated when these patients receive epidural analgesia.²⁷ According to Guglielminotti et al.,²⁸ “More than one-fifth (21%; 95% CI, 14 to 28) of the observed association of neuraxial analgesia with a decreased risk of severe maternal mortality was mediated through the decreased risk of postpartum hemorrhage.” Further studies have shown that neuraxial labor analgesia also decreases

transfusion risk, especially with intrapartum cesarean delivery.²⁹ Therefore, increased access to neuraxial labor analgesia is a dramatic step toward reducing in-hospital maternal morbidity and improving care disparities.³⁰ Future policies should ensure that these services are available and valued by hospital systems in the form of stipends.

Tubal sterilization is the most common method of birth control for females over age 35.³¹ Many of these procedures are postpartum tubal ligations, which are performed immediately after vaginal delivery. The advantages are convenient access to care while in the hospital and the anatomic location of the uterus allowing easy surgical access to the fallopian tubes. These surgeries are more likely to be requested by patients with Medicaid insurance, high-risk pregnancies, higher parity, and those who are a racial or ethnic minority.³² When these requests are not fulfilled, the pregnancy rate in the following year approaches 50%.³³ Based on the very high pregnancy rate with unfulfilled requests, the American College of Obstetricians and Gynecologists has recommended designating postpartum tubal ligations as nonelective procedures.³² Although most hospitals provide postpartum tubal ligations during the delivery admission, this study and the 2011 survey found that stratum I hospitals have more difficulty staffing anesthesia for postpartum tubal ligation, especially at nights and on weekends, while smaller hospitals report less difficulty. Overall, use of spinal, epidural, or general anesthesia was similar, but general

anesthesia was more commonly used for postpartum tubal ligation in smaller hospitals compared to stratum I. Further studies should be undertaken to determine optimal staffing models to reduce the barrier of anesthesia staffing for postpartum tubal ligation surgery.

With higher prevalence of maternal comorbid conditions in the United States³⁰ and more than 80% of maternal mortalities being preventable,³⁴ questions about availability and training for critical obstetric events were incorporated in the 2011 and 2021 surveys. Availability of emergency resources are an important part of patient safety with labor and delivery. For example, massive transfusion protocols for labor and delivery are now widespread, with 92% of respondents reporting access to this lifesaving resource. Even in the smallest hospitals, respondents reported having rapid response teams for labor and delivery. However, many respondents reported they “don’t know” whether their labor and delivery nurses are trained in ACLS. Data support team training for labor and delivery³⁵ and understanding the level of training for nursing staff on labor and delivery is an important component of teamwork.

This study has multiple strengths, including (1) longitudinal survey development and authors; (2) standardized survey questions over 40 yr and expansion to new questions assessing contemporary practice; (3) use of statistical modeling for generalization of survey results to the larger U.S. population; (4) expanded strata to include larger practices; and (5) use of web-push survey methodology to optimize contacts. One limitation is nonresponse bias that we addressed by statistical weighting. The major limitation is low response rate, despite exhaustive searches for chiefs of anesthesiology at hospitals. The searches included SOAP members, obstetric anesthesia division chiefs, obstetric fellowship directors, and personal knowledge of obstetric anesthesiologists by the survey team at individual hospitals that were part of the random sample. The response rate to this survey is consistent with those from similar studies and with widely documented reports of low response in physician surveys.³⁶ We were unable to offer survey participation incentives, which may be one reason why response was not higher. Funding for future surveys tracking obstetric anesthesiology practices will be important in improving provider response.

To our knowledge, this is the only study that reports longitudinal obstetric anesthesia data over 40 yr. Despite stable annual birth rates in the United States, the number of hospitals providing obstetric care decreased by 50% over the last 4 decades. This study describes nonacademic practice and not just academic teaching hospitals. Increased access to neuraxial labor analgesia is a dramatic step toward reducing in-hospital maternal mortality and improving healthcare disparities.

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Competing Interests

Dr. Traynor reports funding from Northwest Anesthesia Seminars (Pasco, Washington) for paid lectures on educational topics in anesthesiology. She received book royalties for the *Manual of Clinical Anesthesiology* (Lippincott Williams & Wilkins, Philadelphia, Pennsylvania). In addition, she served as a legal expert witness in 13 cases from 2022 to 2024 but has no current cases. The other authors declare no competing interests.

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Appendix 1. 40-Year Update of the Obstetric Anesthesia Workforce Survey

Colorado Multiple Institutional Review Board Approval No. 22-0970

Thank you for participating in this survey. Please answer the questions to the best of your ability.

Throughout the survey, the abbreviation L&D will be used to describe the labor and delivery unit.

The Following Questions Pertain to Demographics and Staffing of Your Hospital

1. What role do you provide in your institution?
 - ☐ Obstetric anesthesia fellowship trained anesthesiologist
 - ☐ Anesthesiologist
 - ☐ CRNA
 - ☐ AA
 - ☐ Nurse Manager
 - ☐ Obstetrician
 - ☐ Other (Please specify) _____
2. Who directs the obstetric anesthesia service in your practice?
 - ☐ Obstetric anesthesia fellowship trained anesthesiologist
 - ☐ Anesthesiologist
 - ☐ CRNA
 - ☐ AA
3. How many deliveries occur at your hospital annually?
 - ☐ >4,500
 - ☐ 3,000–4,499
 - ☐ 1,500–2,999
 - ☐ 500–1,499
 - ☐ <500
4. Which of the following best describes your practice?
 - ☐ Academic with an anesthesiology residency program
 - ☐ Academic without an anesthesiology residency program
 - ☐ Private practice with anesthesiology residents
 - ☐ Private practice without anesthesiology residents
 - ☐ Managed care consortium (e.g., Kaiser Permanente)
 - ☐ Military
 - ☐ Indian Health Service Hospital
5. Which of the following best describes your staffing on labor and delivery during weekday daytime hours (7:00 am to 4:00 pm)?
 - ☐ Care team model with physician anesthesiologist supervision with the following types of care providers:
 - ☐ AA
 - ☐ CRNA
 - ☐ Resident/fellow
 - ☐ Combination of above
 - ☐ Physician anesthesiologist(s) personally providing care
 - ☐ CRNA(s) without physician supervision
6. Which of the following best describes your staffing on labor and delivery during nights/weekend hours?
 - ☐ Care team model with physician anesthesiologist supervision with the following types of care providers:
 - ☐ AA
 - ☐ CRNA
 - ☐ Resident/fellow
 - ☐ Combination of above
 - ☐ Physician anesthesiologist(s) personally providing care
 - ☐ CRNA(s) without physician supervision
7. When covering labor and delivery during daytime hours (7:00 am to 4:00 pm), does the physician anesthesiologist have other clinical responsibilities (e.g., operating rooms)?
 - ☐ Yes
 - ☐ No
8. When covering labor and delivery during nights and weekends, does the physician anesthesiologist have other clinical responsibilities (e.g., operating rooms)?
 - ☐ Yes
 - ☐ No
9. Does your hospital offer the anesthesiology group a stipend for in-house obstetric coverage?
 - ☐ Yes
 - ☐ No

- I don't know
- 10. Is your center a high-risk referral center (*i.e.*, Maternal Levels of Care 3 or 4)?
 - Yes
 - No
 - I don't know
- 11. Do you have a means of providing antenatal consultation for high-risk patients prior to their arrival on labor and delivery?
 - Yes, we have an antenatal consultation clinic or service
 - Yes, but only on an ad-hoc basis
 - No
 - I don't know

The Following Questions Pertain to Staffing and Analgesia for Labor

12. Analgesia for Labor: Making your best estimate, about what percentage of your hospital's 2021 maternity cases received the following types of pain relief during labor? Your answers may add up to >100% since many women choose more than one modality.
- _____ None
- _____ IV analgesics, including PCA
- _____ Paracervical block
- _____ Single injection spinal
- _____ Epidural
- _____ Dural puncture epidural (DPE)
- _____ Combined spinal epidural (CSE)
- _____ Nitrous oxide

If you have nitrous oxide available for labor analgesia, is the anesthesiology department responsible for management?

- Yes

If yes, do you bill for the service (*e.g.*, billed as a MAC sedation case)?

- Yes
- No
- I don't know
- No
- I don't know

13. Prior to neuraxial analgesia, does the anesthesiology service require any of the following laboratory studies for a healthy, uncomplicated parturient?

Hemoglobin and hematocrit: Yes_____ No_____ I don't know_____

Platelet count: Yes_____ No_____ I don't know_____

Basic metabolic panel: Yes_____ No_____ I don't know_____

INR or other coagulation studies: Yes_____ No_____ I don't know_____

14. For epidural analgesia, what do you use for maintenance of analgesia?
- Continuous infusion only
 - Continuous infusion plus patient-controlled bolus (PCEA)
 - Programmed intermittent epidural bolus (PIEB) only
 - PIEB plus PCEA
15. Does your hospital offer Trial of Labor after Cesarean (TOLAC)? If no, proceed to question 17.
- No
 - Yes
 - I don't know
16. Does your hospital require the in-house presence of a CRNA or Anesthesiologist for TOLAC patients?
- No
 - Yes, if neuraxial anesthesia is in place
 - Yes, always
 - I don't know
17. Does your hospital require the in-house presence of an obstetrician for TOLAC patients?
- No
 - Yes
 - Don't know

The Next Questions Pertain to Staffing and Anesthesia Technique for Cesarean Delivery and Postpartum Tubal Ligation

18. Anesthesia for ELECTIVE SCHEDULED Cesarean Delivery: Making your best estimate, about what percentage of your hospital's 2021 ELECTIVE SCHEDULED Cesarean Deliveries were done under the following anesthetic techniques? (This must add to 100%.)

_____ Spinal

_____ Epidural

_____ Combined spinal epidural

_____ General

19. Anesthesia for URGENT/EMERGENT Cesarean Delivery: Making your best estimate, about what percentage of your hospital's 2021 URGENT/EMERGENT Cesarean Deliveries were done under the following anesthetic techniques? This must add to 100%.)

_____ Spinal

_____ Epidural from labor

_____ New epidural (*i.e.*, not from labor)

_____ New combined spinal epidural (*i.e.*, not from labor)

_____ General anesthesia

20. Do you routinely administer spinal or epidural morphine to patients undergoing cesarean delivery?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
21. Does your L&D have an ERAS protocol for Cesarean Delivery?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
22. Does your hospital provide **postpartum tubal ligation** surgery during the delivery admission? If no, proceed to question 24.
- ☐ No
 - ☐ Yes
23. If your hospital provides **postpartum tubal ligation** surgeries during the delivery admission, how often does **inadequate anesthesia staffing** affect your ability to perform the surgery?
- ☐ Always
 - ☐ Only nights/weekends
 - ☐ Infrequently (<10%)
 - ☐ Never
24. Making your best estimate, about what percentage of your hospital's 2021 **postpartum tubal ligations** were done under the following anesthetic techniques? This must add to 100%.)
- _____ Spinal
- _____ Epidural from labor
- _____ New epidural (*i.e.*, not from labor)
- _____ New combined spinal epidural (not from labor)
- _____ General anesthesia
25. Does your hospital have a massive transfusion protocol?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
26. Do you have point of care testing (POC) available (*e.g.*, TEG, ROTEM, i-STAT)?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
27. Does your hospital require ACLS or equivalent certification for L&D nurses?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
28. Does your hospital have a rapid response team that responds to critical events on labor and delivery?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
29. Do you have equipment for neuraxial ultrasound on labor and delivery?
- ☐ No
 - ☐ Yes, dedicated to the anesthesia service
 - ☐ Yes, borrowed from the obstetrician or pain service
 - ☐ I don't know
30. Do you have anesthesiology staff proficient with acquisition and interpretation of **transthoracic echo (TTE)** images for labor and delivery?
- ☐ No
 - ☐ Yes
 - ☐ I don't know
31. Do the anesthesiologists have any role in newborn resuscitation?
- ☐ No
 - ☐ Yes
 - ☐ Yes, but only in an emergency
 - ☐ I don't know
32. Making your best estimate, about what percentage of the maternity patients at your hospital are in the racial/ethnic categories listed below?
- White, European descent _____%
- African American _____%
- Hispanic/Latina _____%
- Asian _____%
- Native American _____%
- I don't know
33. Making your best estimate, about what percentage of maternity patients at your hospital are covered by the following insurers (*i.e.*, payor mix)? (This should add to 100%.)
- Medicaid _____%
- Military insurance program _____%
- Private Insurance _____%
- Self-Pay _____%
- Other _____%
- I don't know

The Following Questions Pertain to Emergency Services and Resources

Appendix 2. Survey Response Rates

Year	Hospitals Surveyed, n	Survey Responses, n	Survey Response, %
1981	1,171	617	52*
1991	2,265	747	33
2001	1,300	378	29
2011	1,193	344	29
2021	1,175	284	24

The table summarizes the survey response rates by decade.

*Personal communication between Charles Gibbs, M.D., and Joy Hawkins, M.D. This result is an overestimate of response rate.

Appendix 3. Financial Information and Patient Demographics by Strata

Questions and Responses	Stratum I: >4,499 Births		Stratum II: 3,000–4,499 Births		Stratum III: 1,500–2,999 Births		Stratum IV: 500–1,499 Births		Stratum V: <500 Births		Overall	
	n	%	n	%	n	%	n	%	n	%	n	%
Does your hospital offer the anesthesiology group a stipend for in-house obstetric coverage?												
Yes	22	45.4	19	27.77	50	23.36	160	41.53	55	12.63	307	26.57
Making your best estimate, about what percentage of the maternity patients in your hospital are in the racial/ethnic categories listed below?												
White		40.21		48.54		51.29		55.8		60.35		61.26
African American		23.94		15.13		21.05		16.66		8.54		16.15
Hispanic		26.1		25.79		21.67		19.06		18.41		15.76
Asian		8.64		9.06		4.9		4.51		3.99		3.81
Native American		1.11		1.16		0.75		3.6		6.57		2.5
Making your best estimate, about what percentage of the maternity patients in your hospital are covered by the following insurers?												
Medicaid		31.48		39.08		40.33		47.9		64.21		52.71
Military Insurance Program		1.42		3.77		3.84		2.87		3.7		3.36
Private Insurance		55.42		47.51		43.28		44.09		25.21		36.59
Self-pay		6.45		2.47		4.98		4.18		5.37		4.59
Other		5.22		0.33		3.68		1.13		1.51		1.7

For the first categorical question, the breakdown stratified by stratum is represented as estimated weighted n (estimated weighted percentage of responding hospitals within stratum). For example, it is estimated that 45.4% of stratum hospitals offer the anesthesiology group a stipend. For the remaining two questions, the racial/ethnic categories and insurance coverage are summarized by stratum where each percentage represents the weighted estimated average percentage that was reported by surveyed hospitals.